Q1

$y = mx + 4 \qquad \ldots(\text{i})$

$y^2 = 4x$ tangent $y = mx + \dfrac{a}{m} \Rightarrow y = mx + \dfrac{1}{m} \ \ldots.(\text{ii})$

from (i) and (ii)

$4 = \dfrac{1}{m} \Rightarrow m = \dfrac{1}{4}$

So line $y = \dfrac{1}{4}x + 4$ is also tangent to parabola

$x^2 = 2by$, so solve

$x^2 = 2b\left(\dfrac{x+16}{4}\right)$

$\Rightarrow 2x^2 - bx - 16b = 0 \Rightarrow D = 0$

$\Rightarrow b^2 - 4 \times 2 \times (-16b) = 0$

$\Rightarrow b^2 + 32 \times 4b = 0$

$b = -128,\ b = 0$ (not possible)

Q2

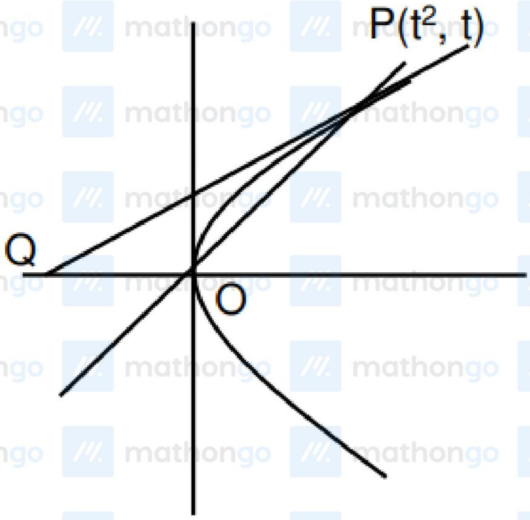
Let point P be $(2t,\ t^2)$ and Q be $(h,\ k)$.

$h = \dfrac{2t}{3},\ k = \dfrac{-2+t^2}{3}$

Hence locus is $3k + 2 = \left(\dfrac{3h}{2}\right)^2$

$\Rightarrow 9x^2 = 12y + 8$

Q3



$2ty = x + t^2$

$Q\left(-t^2, 0\right)$

$\frac{1}{2}\left\|\begin{vmatrix} 0 & 0 & 1 \\ t^2 & t & 1 \\ -t^2 & 0 & 1 \end{vmatrix}\right\| = 4$

$|t|^3 = 8$

$t = \pm 2\,(t > 0)$

$m = \frac{1}{2} = 0.5$

Q4

Let $\left(\frac{1}{2},\ -2\right)$ is $\left(2t^2,\ 4t\right)\ \Rightarrow\ t = \frac{-1}{2}$

Parameter of other end of focal chord is 2

$\Rightarrow$ point is $(8,\ 8)$

$\Rightarrow$ equation of tangent is $8y - 4(x + 8) = 0$

$\Rightarrow\ 2y - x = 8$

Q5

$\because y = x^2 + 7x + 2$

$\therefore \frac{dy}{dx} = 2x + 7$

Slope of tangent parallel to line $y = 3x - 3$

$\therefore 2x + 7 = 3$

$\therefore x = -2$

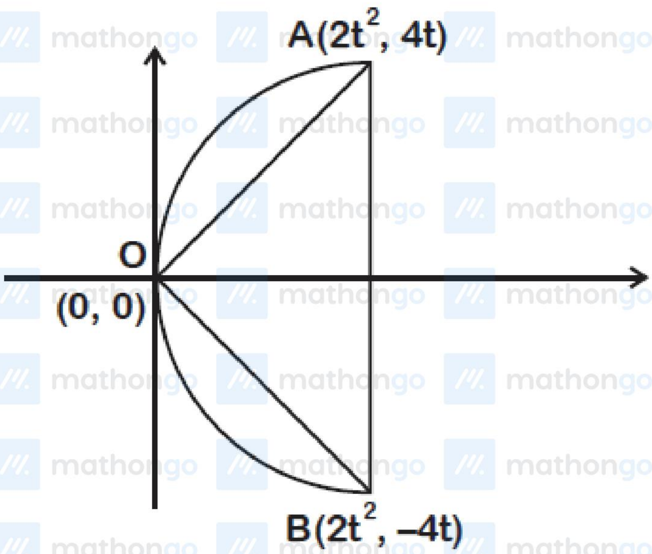
$\therefore$ Point on curve $= (-2, -8)$

Equation of normal at point (-2,-8)

$y+8=-\frac{1}{3}(x+2)$
$x+3y+26=0$

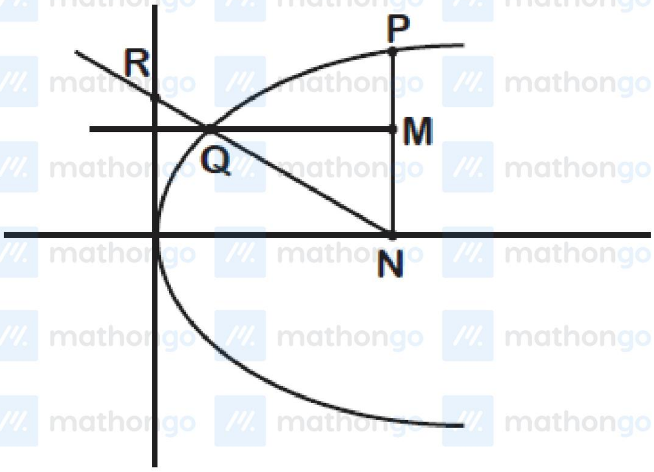
Q6

Let $A=\left(2t^2,4t\right)$
$B\equiv\left(2t^2,4t\right)$ (by symmetry as equilateral triangle)



for equilateral triangle (angle at 0 is 60°)
$\frac{4t}{2t^2}=\frac{1}{\sqrt{3}}\Rightarrow t=2\sqrt{3}$
Area $=\frac{1}{2}\cdot 8(2\sqrt{3})\cdot 2\cdot 24=192\sqrt{3}$

Q7



Let $P\left(at^2, 2at\right)$ where $a=3$

$\Rightarrow N\left(at^2, 0\right) \Rightarrow M\left(at^2, at\right)$

$\because \quad QM \equiv y = at$

So $y^2 = 4ax \quad \Rightarrow \quad x = \frac{at^2}{4}$

$\Rightarrow Q\left(\frac{at^2}{4}, at\right)$

$\Rightarrow \quad QN \equiv y = \frac{-4}{3t}\left(x - at^2\right)$

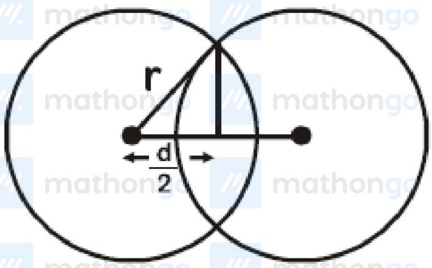
$\because \quad QN$ passes through $\left(0, \frac{4}{3}\right)$, then

$\frac{4}{3} = -\frac{4}{3t}\left(-at^2\right) \quad \Rightarrow \quad$ at $= 1 \quad \Rightarrow \quad t = \frac{1}{3}$

Now, $\mathrm{MQ} = \frac{3}{4}\mathrm{at}^2 = \frac{1}{4}$ and $\mathrm{PN} = 2$ at $= 2$

Q8

Let the distance between centres be "d'.



Length of common chord $= 2\sqrt{r^2 - \left(\frac{d}{2}\right)^2} = 4$

$\Rightarrow r^2 - \frac{d^2}{4} = 4 \quad (\text{ given } r = 2\sqrt{5})$

$\Rightarrow \frac{d^2}{4} = 16$

$\Rightarrow d = 8$

Q9

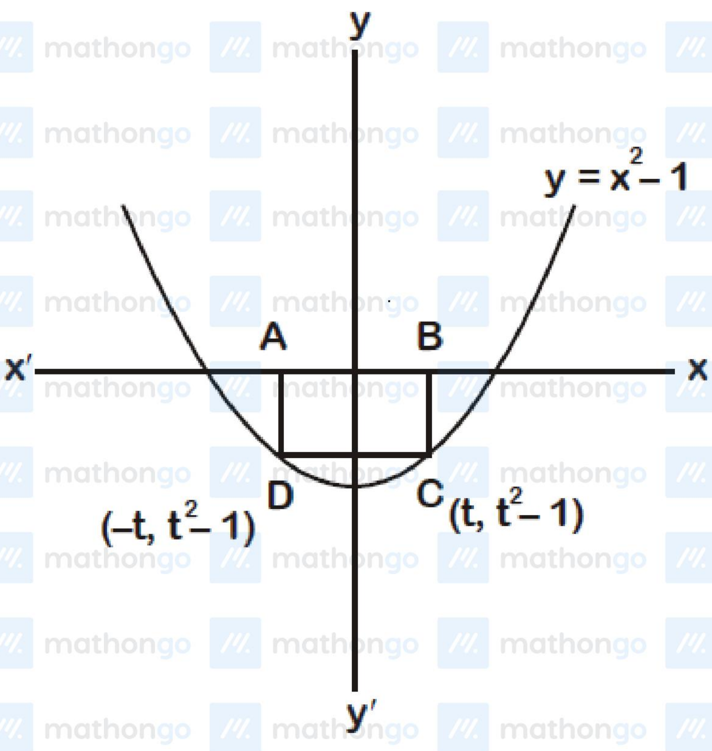
Area of rectangle ABCD

$A = \left|2t \cdot \left(t^2 - 1\right)\right|$

$A = \left|2t^3 - 2t\right|$

$\therefore \quad \frac{\mathrm{dA}}{\mathrm{dt}} = \left|6\mathrm{t}^2 - 2\right|$

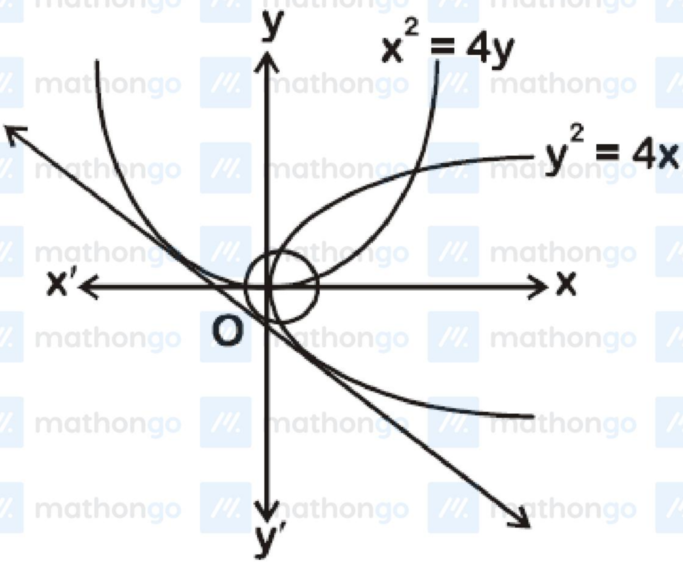
For maximum area $\frac{\mathrm{dA}}{\mathrm{dt}} = 0 \Rightarrow \mathrm{t} = \pm\frac{1}{\sqrt{3}}$



$\therefore \quad$ Maximum area $= \left|\frac{2}{3\sqrt{3}} - \frac{2}{\sqrt{3}}\right| = \frac{4}{3\sqrt{3}}$

Q10

Equation tangent to parabola $y^2 = 4x$
with given slope m is :



$y = mx + \frac{1}{m}$
$\because$ Line (i) is tangent to $x^2 = 4y$
$\therefore \quad x^2 = 4mx + \frac{4}{m}$
$\Rightarrow \quad mx^2 - 4m^2x - 4 = 0$
For tangent : $16\mathrm{m}^4 + 16\mathrm{m} = 0$
$16m\left(m^3 + 1\right) = 0$
$m = 0, -1$
$\therefore \quad$ Equation tangent : $x + y + 1 = 0$
It is tangent to circle $x^2 + y^2 = c^2$
$\Rightarrow \quad c = \frac{1}{\sqrt{2}}$

Q11

$y = (x-1)(x-2)$, this curve intersects the x -axis at $\mathrm{A}(1,0)$ and $\mathrm{B}(2,0)$
$\therefore \frac{dy}{dx} = 2x - 3; \frac{dy}{dx(x=1)} = -1$ and $\frac{dy}{dx(x=2)} = 1$
Equation of tangent at $\mathrm{A}(1,0)$;
$y = -1(x-1)$
$\Rightarrow x + y = 1$
and equation of tangent at $\mathrm{B}(2,0)$
$y = 1(x-2)$

$\Rightarrow \quad x - y = 2$
So $a = 1$ and $b = 2$
$\Rightarrow \frac{a}{b} = 0.5$

Q12

$L_1 : y = m_1(x+1) + \frac{1}{m_1}$
$L_2 : y = m_2(x+2) + \frac{2}{m_2}$
If L_1 and L_2 intersects at (h, k) then
$m_1^2(h+1) - km_1 + 1 = 0$
$m_2^2(h+2) - km_2 + 1 = 0$
$\because \quad m_2 = -\frac{1}{m_1}$
$\Rightarrow \quad m_1^2 + km_1 + (h+2) = 0$
from (1) and (3)
$\frac{h+1}{1} = \frac{-k}{k} = \frac{1}{h+2}$
$\Rightarrow h + 3 = 0$
$\Rightarrow x + 3 = 0$

Q1 JEE Main 2020 - 7 Jan (Morning)

If $y = mx + 4$ is a tangent to both the parabolas $y^2 = 4x$ and $x^2 = 2by$. Then value of b is

(A) -64
(B) -32
(C) -128
(D) 16

Q2 JEE Main 2020 - 8 Jan (Morning)

The locus of a point which divides the line segment joining the point $(0, \ -1)$ and a point on the parabola, $x^2 = 4y$, internally in the ratio $1 : 2$, is :

(A) $9x^2 = 3y + 2$
(B) $9x^2 = 12y + 8$
(C) $9y^2 = 12x + 8$
(D) $9y^2 = 3x + 2$

Q3 JEE Main 2020 - 8 Jan (Evening)

Let a line $y = mx \ (m > 0)$ intersect the parabola, $y^2 = x$ at a point P, other than the origin. Let the tangent to it at P meet the x-axis at the point Q. If area $(\Delta OPQ) = 4$ sq. units, then m is equal to

Q4 JEE Main 2020 - 9 Jan (Evening)

If one end of a focal chord AB of the parabola $y^2 = 8x$ is at $A\left(\frac{1}{2}, \ -2\right)$, then the equation of the tangent to it at B is :

(A) $x - 2y + 8 = 0$
(B) $x + 2y + 8 = 0$
(C) $2x - y - 24 = 0$
(D) $2x + y - 24 = 0$

Q5 JEE Main 2020 - 2 Sep (Morning)

Let $P(h, \quad k)$ be a point on the curve $y = x^2 + 7x + 2$, nearest to the line, $y = 3x - 3$ Then the equation of the normal to the curve at P is :

(A) $x - 3y - 11 = 0$

(B) $x-3y+22=0$
(C) $x+3y-62=0$
(D) $x+3y+26=0$

Q6 JEE Main 2020 - 2 Sep (Evening)

The area (in sq. units) of an equilateral triangle inscribed in the parabola $y^2=8x$, with one of its vertices on the vertex of this parabola, is :

(A) $64\sqrt{3}$
(B) $256\sqrt{3}$
(C) $128\sqrt{3}$
(D) $192\sqrt{3}$

Q7 JEE Main 2020 - 3 Sep (Morning)

Let P be a point on the parabola, $y^2=12x$ and N be the foot of the perpendicular drawn from P on the axis of the parabola. A line is now drawn through the mid-point M of PN, parallel to its axis which meets the parabola at Q. If the y -intercept of the line NQ is $\frac{4}{3}$, then :

(A) $\mathrm{MQ}=\frac{1}{4}$
(B) $\mathrm{PN}=3$
(C) $\mathrm{PN}=4$
(D) $\mathrm{MQ}=\frac{1}{3}$

Q8 JEE Main 2020 - 3 Sep (Evening)

Let the latus ractum of the parabola $y^2=4x$ be the common chord to the circles C_1 and C_2 each of them having radius $2\sqrt{5}$. Then, the distance between the centres of the circles C_1 and C_2 is :

(A) 8
(B) 12
(C) $8\sqrt{5}$
(D) $4\sqrt{5}$

Q9 JEE Main 2020 - 4 Sep (Evening)

The area (in sq. units) of the largest rectangle ABCD whose vertices A and B lie on the x -axis and vertices C and D lie on the parabola, $y=x^2-1$ below the x -axis, is :

(A) $\frac{4}{3\sqrt{3}}$
(B) $\frac{1}{3\sqrt{3}}$
(C) $\frac{2}{3\sqrt{3}}$
(D) $\frac{4}{3}$

Q10 JEE Main 2020 - 5 Sep (Morning)

If the common tangent to the parabolas, $y^2 = 4x$ and $x^2 = 4y$ also touches the circle, $x^2 + y^2 = c^2$, then c is equal to:

(A) $\frac{1}{2}$
(B) $\frac{1}{4}$
(C) $\frac{1}{2\sqrt{2}}$
(D) $\frac{1}{\sqrt{2}}$

Q11 JEE Main 2020 - 5 Sep (Evening)

If the lines $x + y = a$ and $x - y = b$ touch the curve $y = x^2 - 3x + 2$ at the points where the curve intersects the x-axis, then $\frac{a}{b}$ is equal to

Q12 JEE Main 2020 - 6 Sep (Morning)

Let L_1 be a tangent to the parabola $y^2 = 4(x+1)$ and L_2 be a tangent to the parabola $y^2 = 8(x+2)$ such that L_1 and L_2 intersect at right angles. Then L_1 and L_2 meet on the straight line:

(A) $2x + 1 = 0$
(B) $x + 3 = 0$
(C) $x + 2y = 0$
(D) $x + 2 = 0$

Answer Key

Q1 (C)	**Q2** (B)	**Q3** (0.5)	**Q4** (A)
Q5 (D)	**Q6** (D)	**Q7** (A)	**Q8** (A)
Q9 (A)	**Q10** (D)	**Q11** (0.5)	**Q12** (B)